Exam 3 Study Guide

AC Circuit Analysis — Sadiku Chapters 9–12 and 14

Phasors · AC Analysis · AC Power · Three-Phase · Frequency Response

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How to use this. Every worked problem below starts from a schematic, because that is how the exam problems are posed. Cover the solution, redraw the circuit in the frequency domain yourself, then check. The five chapters break down as: Ch. 9 (build the phasor circuit), Ch. 10 (apply the DC techniques to it), Ch. 11 (extract power from it), Ch. 12 (three of them at once), Ch. 14 (sweep ω). Section 9 at the end is a full-length practice exam with an answer key.

1 Reference Sheet

1.1 Phasor transform

A sinusoid must be written as a *cosine with positive amplitude* before you take its phasor. Two identities do all the work:

$$\sin(\omega t) = \cos(\omega t - 90^\circ), \qquad -\cos(\omega t) = \cos(\omega t \pm 180^\circ)$$

Time domain	Phasor domain
$v(t) = V_m \cos(\omega t + \phi)$	$\mathbf{V} = V_m \angle \phi$
$\dfrac{dv}{dt}$	$\mathrm{j}\omega \mathbf{V}$
$\int v\, dt$	$\dfrac{\mathbf{V}}{\mathrm{j}\omega}$

1.2 Impedance and admittance

Element	Impedance $\mathbf{Z}$	Admittance $\mathbf{Y}$	Phase of v rel. to i
Resistor R	R	$1/R$	in phase
Inductor L	$\mathrm{j}\omega L$	$1/(\mathrm{j}\omega L)$	v leads i by 90°
Capacitor C	$1/(\mathrm{j}\omega C) = -\mathrm{j}/(\omega C)$	$\mathrm{j}\omega C$	v lags i by 90°

$\mathbf{Z} = R + \mathrm{j}X$ where $X > 0$ is inductive and $X < 0$ is capacitive. $\mathbf{Y} = G + \mathrm{j}B$, and note that $G \neq 1/R$ in general.

Memory aid: ELI the **ICE** man. In an inductor (L), voltage E leads current I. In a capacitor (C), current I leads voltage E.

1.3 AC power (Ch. 11)

Let $\mathbf{V} = V_m \angle \theta_v$ and $\mathbf{I} = I_m \angle \theta_i$ be *amplitude* phasors, and $\theta = \theta_v - \theta_i$.

Quantity	Formula	Units
RMS value	$V_{\text{rms}} = V_m/\sqrt{2}$ (sinusoids only)	V
Average (real) power	$P = \frac{1}{2}V_m I_m \cos\theta = V_{\text{rms}} I_{\text{rms}} \cos\theta$	W
Reactive power	$Q = \frac{1}{2}V_m I_m \sin\theta = V_{\text{rms}} I_{\text{rms}} \sin\theta$	VAR
Complex power	$\mathbf{S} = \frac{1}{2}\mathbf{V}\mathbf{I}^* = \mathbf{V}_{\text{rms}}\mathbf{I}^*_{\text{rms}} = P + \text{j}Q$	VA
Apparent power	$S = \|\mathbf{S}\| = V_{\text{rms}} I_{\text{rms}}$	VA
Power factor	$\text{pf} = \cos\theta = P/S$	—
Also	$\mathbf{S} = I^2_{\text{rms}}\mathbf{Z} = V^2_{\text{rms}}/\mathbf{Z}^*$	VA

Resistors absorb only P. Inductors absorb only $+Q$. Capacitors absorb only $-Q$. Lagging pf $\Rightarrow$ inductive load, $\theta > 0$, current lags voltage. Leading pf $\Rightarrow$ capacitive load, $\theta < 0$.

Maximum average power transfer: $\mathbf{Z}_L = \mathbf{Z}^*_{\text{Th}} = R_{\text{Th}} - \text{j}X_{\text{Th}}$, giving

$$P_{\max} = \frac{|\mathbf{V}_{\text{Th}}|^2}{8R_{\text{Th}}} \quad \text{(amplitude phasor)} \qquad P_{\max} = \frac{|\mathbf{V}_{\text{Th,rms}}|^2}{4R_{\text{Th}}} \quad \text{(rms phasor)}$$

If $\mathbf{Z}_L$ is restricted to be purely resistive, then $R_L = |\mathbf{Z}_{\text{Th}}| = \sqrt{R^2_{\text{Th}} + X^2_{\text{Th}}}$.

Power factor correction (lagging $\to$ better lagging), shunt capacitor:

$$Q_C = P(\tan\theta_1 - \tan\theta_2), \qquad C = \frac{Q_C}{\omega V^2_{\text{rms}}}$$

1.4 Three-phase (Ch. 12)

Positive (abc) sequence: $\mathbf{V}_{an} = V_p\angle 0°$, $\mathbf{V}_{bn} = V_p\angle{-120°}$, $\mathbf{V}_{cn} = V_p\angle{+120°}$.

	Y connection	Δ connection
Line vs. phase voltage	$V_L = \sqrt{3}\,V_p$, $\mathbf{V}_{ab}$ leads $\mathbf{V}_{an}$ by 30°	$V_L = V_p$
Line vs. phase current	$I_L = I_p$	$I_L = \sqrt{3}\,I_p$, $\mathbf{I}_a$ lags $\mathbf{I}_{AB}$ by 30°

Balanced total power, *same formula for both connections*:

$$P_T = \sqrt{3}\,V_L I_L \cos\theta, \qquad Q_T = \sqrt{3}\,V_L I_L \sin\theta, \qquad S_T = \sqrt{3}\,V_L I_L$$

where θ is the angle of the *per-phase load impedance*, not the angle of V_L or I_L. Instantaneous total power in a balanced system is constant.

Conversion: $\mathbf{Z}_Y = \mathbf{Z}_\Delta/3$. Convert everything to Y and analyze one phase, then use symmetry ($\mp 120°$) for the other two.

Two-wattmeter method:

$$P_1 = V_L I_L \cos(\theta + 30°), \quad P_2 = V_L I_L \cos(\theta - 30°)$$

$$P_T = P_1 + P_2, \qquad Q_T = \sqrt{3}\,(P_2 - P_1), \qquad \theta = \tan^{-1}\left[\sqrt{3}\,\frac{P_2 - P_1}{P_2 + P_1}\right]$$

1.5 Frequency response (Ch. 14)

$$\mathbf{H}(\omega) = \frac{\mathbf{Y}(\omega)}{\mathbf{X}(\omega)}, \qquad H_{\mathrm{dB}} = 20\log_{10}|\mathbf{H}| \;\; \text{(for gain)}, \qquad 10\log_{10}(P_2/P_1) \;\; \text{(for power)}$$

	ω_0	Q	B	$\omega_{1,2}$ (high Q)
Series RLC	$\dfrac{1}{\sqrt{LC}}$	$\dfrac{\omega_0 L}{R} = \dfrac{1}{\omega_0 RC}$	$\dfrac{R}{L} = \dfrac{\omega_0}{Q}$	$\omega_0 \mp B/2$
Parallel RLC	$\dfrac{1}{\sqrt{LC}}$	$\dfrac{R}{\omega_0 L} = \omega_0 RC$	$\dfrac{1}{RC} = \dfrac{\omega_0}{Q}$	$\omega_0 \mp B/2$

Always: $\omega_0 = \sqrt{\omega_1\omega_2}$ and $B = \omega_2 - \omega_1$ and $Q = \omega_0/B$. Exact half-power frequencies (series): $\omega_{1,2} = \mp\dfrac{R}{2L} + \sqrt{\left(\dfrac{R}{2L}\right)^2 + \dfrac{1}{LC}}$. High-$Q$ means $Q \geq 10$.

Bode straight-line rules. Put $\mathbf{H}$ in standard form so every factor reads $(1 + \mathrm{j}\omega/z)$ or $(1 + \mathrm{j}\omega/p)$:

Factor	Magnitude slope after corner	Phase contribution
K (constant)	0, at $20\log_{10}|K|$	0° (180° if $K < 0$)
$(\mathrm{j}\omega)^N$	$+20N$ dB/dec, all ω	$+90N^\circ$
$(\mathrm{j}\omega)^{-N}$	$-20N$ dB/dec, all ω	$-90N^\circ$
$(1 + \mathrm{j}\omega/z)$ zero	$+20$ dB/dec above $\omega = z$	$0^\circ \to +90^\circ$ over $z/10$ to $10z$
$(1 + \mathrm{j}\omega/p)^{-1}$ pole	-20 dB/dec above $\omega = p$	$0^\circ \to -90^\circ$ over $p/10$ to $10p$

A quadratic pole/zero doubles those numbers (± 40 dB/dec, $\pm 180^\circ$).

Scaling. With magnitude factor K_m and frequency factor K_f:

$$R' = K_m R, \qquad L' = \frac{K_m}{K_f} L, \qquad C' = \frac{C}{K_m K_f}$$

Q is unchanged by both. ω_0 and B scale by K_f and are unaffected by K_m.

2 Analysis Methods Refresher

These are the Chapter 3–4 techniques. They are worth re-reading because in Chapters 10–12 you apply them unchanged, just with complex numbers. Nothing about the *method* changes when you move to AC — only the arithmetic does.

2.1 Kirchhoff's laws and sign conventions

KCL (current law). The algebraic sum of currents entering any node is zero. Equivalently, $\sum I_{\text{in}} = \sum I_{\text{out}}$. This follows from conservation of charge, and it holds for a *supernode* (any closed surface enclosing several nodes) just as well as for a single node.

KVL (voltage law). The algebraic sum of voltages around any closed loop is zero. This follows from conservation of energy.

Passive sign convention. Assign the current entering the + terminal of an element. Then $p = vi$ is power *absorbed*. A positive result means the element absorbs; negative means it delivers. Resistors always absorb. Sources can do either.

Walking a KVL loop. Pick a direction and stay with it. When you enter an element at its $-$ terminal and leave at $+$, that is a voltage *rise* (write $-V$ if you are summing drops). Enter at $+$ and leave at $-$: a *drop* (write $+V$). For a resistor, the + terminal is wherever your assumed current enters.

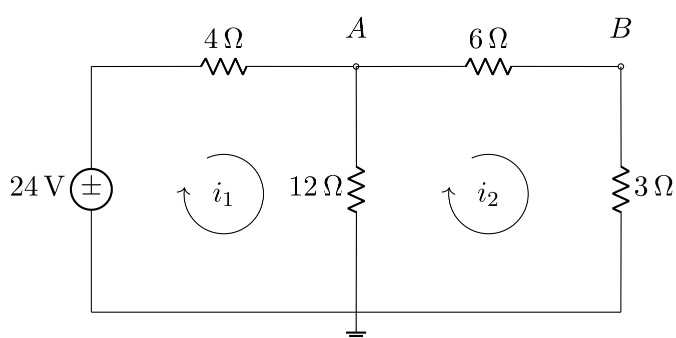
The two you will use constantly:

$$\text{Voltage divider: } v_k = v_s \cdot \frac{R_k}{R_1 + R_2 + \cdots} \qquad \text{Current divider (two branches): } i_1 = i_s \cdot \frac{R_2}{R_1 + R_2}$$

Note the current divider puts the *other* resistor in the numerator. In AC these become $\mathbf{Z}_k / \sum \mathbf{Z}$ and $\mathbf{Z}_2/(\mathbf{Z}_1 + \mathbf{Z}_2)$.

2.2 One circuit, four methods

The same circuit is solved below by nodal analysis, mesh analysis, and Thévenin reduction. Getting the same answer three ways is the best way to build confidence, and it is exactly the cross-check you should run on an exam when you have time left over.



Method 1 — Nodal analysis

Procedure.

1. Pick a reference node (ground). Choose the one with the most branches attached.
2. Label the remaining node voltages $v_1, v_2, \dots$ relative to ground.
3. Write KCL at each non-reference node. Express every branch current as a node-voltage difference over a resistance: $i = \dfrac{v_{\text{near}} - v_{\text{far}}}{R}$, always *leaving* the node.
4. Solve. The number of equations is (number of nodes − 1).

Here node B is not independent — the $6\,\Omega$ and $3\,\Omega$ are simply in series, so treat them as $9\,\Omega$ and there is one unknown, v_A. KCL at A, writing all currents as leaving:

$$\frac{v_A - 24}{4} + \frac{v_A}{12} + \frac{v_A}{9} = 0$$

Multiply through by 36:

$$9(v_A - 24) + 3v_A + 4v_A = 0 \quad \Rightarrow \quad 16v_A = 216 \quad \Rightarrow \quad \boxed{v_A = 13.5\,\text{V}}$$

Then $i_1 = (24 - 13.5)/4 = 2.625\,\text{A}$ and $i_2 = 13.5/9 = 1.5\,\text{A}$.

Supernode. If a voltage source sits between two non-reference nodes, you cannot write its current. Draw a surface around both nodes, write one KCL for the whole surface, and supply the missing equation from the source itself: $v_1 - v_2 = V_{\text{source}}$. See §4.1 for the AC version.

Method 2 — Mesh analysis

Procedure.

1. Confirm the circuit is planar (no crossing wires).
2. Assign a mesh current to each window, all in the *same* direction (clockwise by convention). This consistency is what makes the sign pattern below work.
3. Write KVL around each mesh. In a shared branch, the net current is the difference of the two mesh currents.
4. Solve. The number of equations is the number of windows.

Mesh 1 (clockwise, starting at the source):

$$-24 + 4i_1 + 12(i_1 - i_2) = 0 \quad \Rightarrow \quad 16i_1 - 12i_2 = 24$$

Mesh 2 (clockwise):

$$12(i_2 - i_1) + 6i_2 + 3i_2 = 0 \quad \Rightarrow \quad -12i_1 + 21i_2 = 0$$

From the second equation $i_2 = \frac{12}{21} i_1 = 0.5714\, i_1$. Substituting:

$$16i_1 - 6.857i_1 = 24 \quad \Rightarrow \quad \boxed{i_1 = 2.625\,\text{A}, \quad i_2 = 1.5\,\text{A}}$$

and $v_A = 12(i_1 - i_2) = 12(1.125) = 13.5\,\text{V}$, matching Method 1. ✓

The mesh matrix has a pattern. For a circuit with only independent voltage sources and all mesh currents clockwise:

$$\begin{bmatrix} \sum R_{\text{mesh 1}} & -R_{\text{shared 12}} \\ -R_{\text{shared 12}} & \sum R_{\text{mesh 2}} \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} \sum V_{\text{rises, mesh 1}} \\ \sum V_{\text{rises, mesh 2}} \end{bmatrix}$$

Diagonal entries are positive sums; off-diagonal entries are negative shared impedances; the matrix is symmetric. Above: $\begin{bmatrix} 16 & -12 \\ -12 & 21 \end{bmatrix}$. If your matrix is not symmetric, you have a sign error or a dependent source.

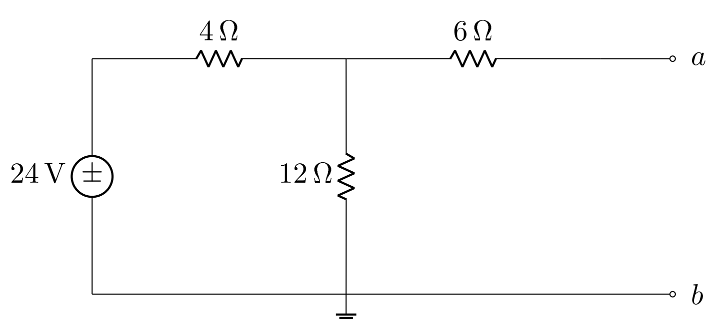
Supermesh. If a *current* source sits in a shared branch, its voltage is unknown, so you cannot write KVL through it. Combine the two meshes into one supermesh, write KVL around the outside, and supply the missing equation from the source: $i_2 - i_1 = I_{\text{source}}$ (sign set by the source's arrow). See §4.2.

Which method? Count the equations each would need. Fewer nodes than meshes ⇒ nodal. Fewer meshes than nodes ⇒ mesh. Current sources favor nodal (they hand you a branch current directly); voltage sources favor mesh. If the circuit is non-planar, mesh analysis is off the table entirely.

Method 3 — Thévenin equivalent

Any linear two-terminal network can be replaced, as seen from those terminals, by a single voltage source V_{Th} in series with a single resistance R_{Th}. This is the right tool when one element is being varied or is the only thing you care about.

Say we want the current through the $3\,\Omega$ resistor. Remove it and look back into terminals a–b:



Step 1 — V_{Th} (open-circuit voltage). With a–b open, no current flows in the $6\,\Omega$ resistor, so it drops zero volts. V_{Th} is therefore just the voltage across the $12\,\Omega$, by voltage division:

$$V_{\text{Th}} = 24 \cdot \frac{12}{4+12} = 18\,\text{V}$$

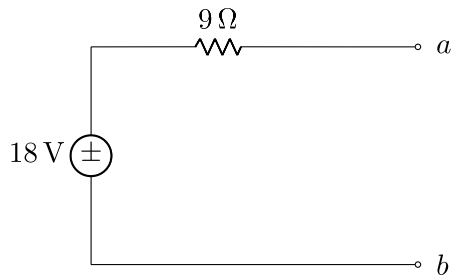
Step 2 — R_{Th}. Kill the independent source (voltage source → short, current source → open) and look in from a–b:

$$R_{\text{Th}} = 6 + (4 \parallel 12) = 6 + \frac{(4)(12)}{16} = 6 + 3 = 9\,\Omega$$

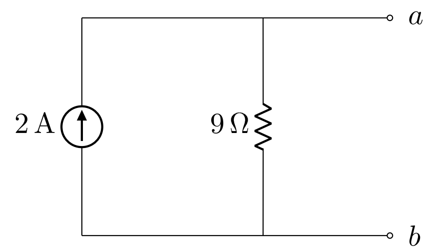
Step 3 — reattach the load.

$$i_{3\Omega} = \frac{V_{\text{Th}}}{R_{\text{Th}} + R_L} = \frac{18}{9+3} = 1.5\,\text{A}$$

which is exactly the i_2 found by mesh analysis. ✓



Thévenin equivalent



Norton equivalent

Method 4 — Norton equivalent and source transformation

The Norton form is a current source I_N in parallel with R_N, and the two equivalents are related by

$$I_N = \frac{V_{\text{Th}}}{R_{\text{Th}}}, \qquad R_N = R_{\text{Th}}$$

Here $I_N = 18/9 = 2\,\text{A}$ and $R_N = 9\,\Omega$.

Verify I_N independently by shorting a–b in the original circuit: the $6\,\Omega$ then parallels the $12\,\Omega$ to give $4\,\Omega$, the source drives $24/(4+4) = 3\,\text{A}$, node A sits at $3(4) = 12\,\text{V}$, and $i_{sc} = 12/6 = 2\,\text{A}$. ✓

Three ways to get R_{Th} — pick based on what the circuit contains:

Circuit contains	Method
Independent sources only	Kill all sources, combine resistances by inspection
Dependent sources present	$R_{\text{Th}} = V_{oc}/I_{sc}$, or apply a 1 V test source at a–b with independent sources killed and compute $R_{\text{Th}} = 1/i_{\text{test}}$
Dependent sources only (no independent)	$V_{\text{Th}} = 0$; must use the test-source method

Source transformation is the same idea applied mid-circuit to simplify as you go: a voltage source v_s in series with R is interchangeable with a current source v_s/R in parallel with R. Watch the polarity — the current source arrow points toward the source's + terminal.

2.3 Superposition

For a linear circuit with multiple independent sources, the response is the sum of the responses to each source acting alone.

1. Leave one independent source active; **short** the other voltage sources and **open** the other current sources.
2. Solve for the quantity of interest.
3. Repeat for each source, then add the individual contributions algebraically.

Never turn off a dependent source. It stays in every sub-circuit.

Superposition does not work for power. $P = i^2 R$ is nonlinear, so you must find the total current first and then compute power. Adding $i_1^2 R + i_2^2 R$ gives the wrong answer.

2.4 Maximum power transfer

For a purely resistive DC network, the load drawing maximum power is $R_L = R_{\text{Th}}$, delivering

$$P_{\text{max}} = \frac{V_{\text{Th}}^2}{4R_{\text{Th}}}$$

For the example above, $R_L = 9\,\Omega$ gives $P_{\text{max}} = 18^2/36 = 9\,\text{W}$. The AC generalization ($\mathbf{Z}_L = \mathbf{Z}_{\text{Th}}^*$) is in §5.3.

What carries over to AC without modification: KCL, KVL, nodal, mesh, supernode, supermesh, superposition, source transformation, Thévenin, Norton, voltage and current division, and series/parallel combination.

What changes: $R \to \mathbf{Z}$, real arithmetic $\to$ complex arithmetic, and $R_L = R_{\text{Th}} \to \mathbf{Z}_L = \mathbf{Z}_{\text{Th}}^*$ for maximum power. That is the entire list.

3 Chapter 9 — Sinusoids and Phasors

3.1 What you have to be able to do

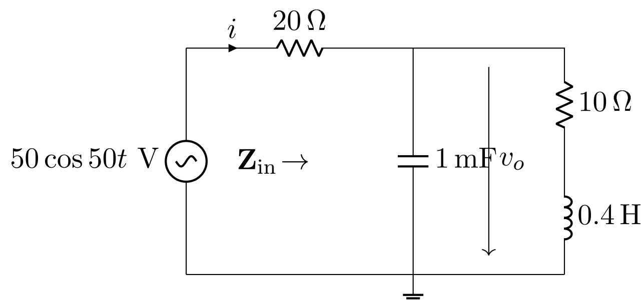
1. Read V_m, ω, f, T, and ϕ off a sinusoid, and determine lead/lag between two sinusoids of the *same* frequency.
2. Redraw a given schematic in the frequency domain: sources become phasors, L becomes $\mathrm{j}\omega L$, C becomes $1/\mathrm{j}\omega C$, resistors stay put.
3. Combine impedances in series and parallel, including delta–wye when needed.
4. Convert back to the time domain at the end.

The single most common exam mistake is forgetting that ω must come from the source before you can convert L and C. Write $\omega =$ ________ rad/s in the corner of your page as step zero. If two sources have different ω, you *cannot* put them in the same phasor circuit — that is a superposition problem (see §4.3).

3.2 Worked problem 9.1 — Input impedance of a ladder network

Problem.

For the circuit below, find the input impedance $\mathbf{Z}_{\text{in}}$ seen by the source, then find $i(t)$ and $v_o(t)$.

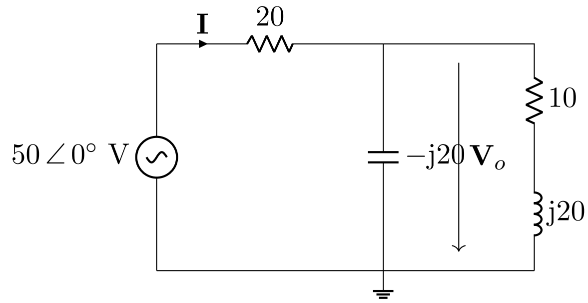


Solution.

Step 0 — extract ω. The source is $50\cos 50t$, so $\omega = 50$ rad/s and the phasor source is $\mathbf{V}_s = 50\angle 0°$ V.

Step 1 — convert to the frequency domain.

$$1\,\text{mF}: \ \frac{1}{\mathrm{j}\omega C} = \frac{1}{\mathrm{j}(50)(10^{-3})} = -\mathrm{j}20\ \Omega \qquad 0.4\,\text{H}: \ \mathrm{j}\omega L = \mathrm{j}(50)(0.4) = \mathrm{j}20\ \Omega$$



Step 2 — combine. The capacitor is in parallel with the series RL branch:

$$\mathbf{Z}_p = \frac{(-\mathrm{j}20)(10+\mathrm{j}20)}{(-\mathrm{j}20)+(10+\mathrm{j}20)} = \frac{400-\mathrm{j}200}{10} = 40-\mathrm{j}20\ \Omega = 44.72\angle-26.57^\circ\ \Omega$$

$$\mathbf{Z}_{\text{in}} = 20+\mathbf{Z}_p = 60-\mathrm{j}20\ \Omega = 63.25\angle-18.43^\circ\ \Omega$$

Step 3 — solve.

$$\mathbf{I} = \frac{50\angle 0^\circ}{63.25\angle-18.43^\circ} = 0.791\angle 18.43^\circ\ \text{A}$$

$$\mathbf{V}_o = \mathbf{I}\,\mathbf{Z}_p = 0.791\angle 18.43^\circ \cdot 44.72\angle-26.57^\circ = 35.36\angle-8.13^\circ\ \text{V}$$

Step 4 — back to time domain (put ω back in):

$$\boxed{i(t) = 0.791\cos(50t+18.43^\circ)\ \text{A}} \qquad \boxed{v_o(t) = 35.36\cos(50t-8.13^\circ)\ \text{V}}$$

Sanity check. $\mathbf{Z}_{\text{in}}$ has a negative reactance, so the network looks capacitive overall and the current should *lead* the source voltage. It does, by 18.43°. Also verify KCL at the top node: $\mathbf{I}_C = \mathbf{V}_o/(-\mathrm{j}20) = 1.768\angle 81.87^\circ$ A and $\mathbf{I}_{RL} = \mathbf{V}_o/(10+\mathrm{j}20) = 1.581\angle-71.57^\circ$ A. Their sum is $(0.25+\mathrm{j}1.75)+(0.5-\mathrm{j}1.5) = 0.75+\mathrm{j}0.25 = 0.791\angle 18.43^\circ$ A. ✓

3.3 Worked problem 9.2 — Working backwards from a branch quantity

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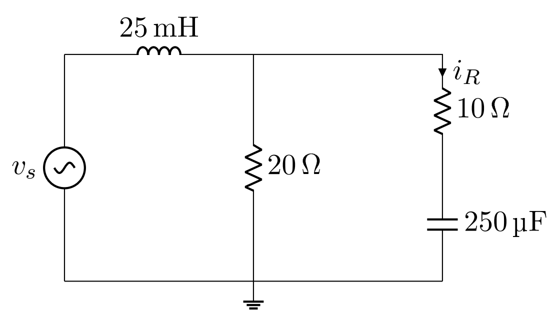
Problem.

In the circuit below, the current through the $10\,\Omega$ resistor is $i_R(t) = 2\cos(200t)$ A. Find $v_s(t)$.



Solution. $\omega = 200$ rad/s.

$$\mathrm{j}\omega L = \mathrm{j}(200)(0.025) = \mathrm{j}5\ \Omega \qquad \frac{1}{\mathrm{j}\omega C} = \frac{1}{\mathrm{j}(200)(250\times 10^{-6})} = -\mathrm{j}20\ \Omega$$

Start from the known branch and walk outward. With $\mathbf{I}_R = 2\angle 0°$ A,

$$\mathbf{V}_{\mathrm{mid}} = \mathbf{I}_R(10-\mathrm{j}20) = 2\angle 0° \cdot 22.36\angle -63.43° = 44.72\angle -63.43°\ \mathrm{V}$$

That same voltage sits across the $20\,\Omega$ resistor, so

$$\mathbf{I}_{20} = \frac{44.72\angle -63.43°}{20} = 2.236\angle -63.43°\ \mathrm{A}$$

KCL gives the source current:

$$\mathbf{I}_s = \mathbf{I}_R + \mathbf{I}_{20} = 2 + (1-\mathrm{j}2) = 3-\mathrm{j}2 = 3.606\angle -33.69°\ \mathrm{A}$$

KVL around the source loop:

$$\mathbf{V}_s = \mathbf{I}_s(\mathrm{j}5) + \mathbf{V}_{\mathrm{mid}} = 3.606\angle -33.69° \cdot 5\angle 90° + (20-\mathrm{j}40)$$

$$= (10+\mathrm{j}15) + (20-\mathrm{j}40) = 30-\mathrm{j}25 = 39.05\angle -39.81°\ \mathrm{V}$$

$$\boxed{v_s(t) = 39.05\cos(200t - 39.81°)\ \mathrm{V}}$$

Why this pattern matters. Any problem phrased "given that x in this branch is . . . , find y" is solved this way: never set up simultaneous equations. Convert the known quantity to a phasor, propagate it with Ohm's law and KCL/KVL, and stop when you reach the unknown.

3.4 Worked problem 9.3 — Series–parallel with a nested branch

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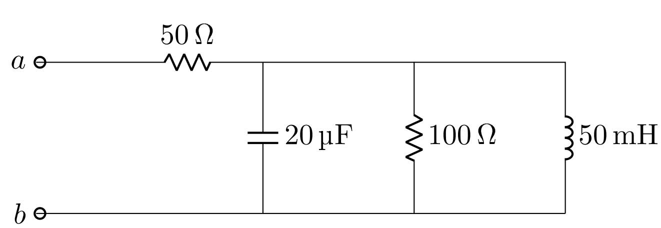
Problem.

Find the input impedance at terminals a–b at $\omega = 1000\,\mathrm{rad\,s^{-1}}$.



Solution.

$$\frac{1}{\mathrm{j}\omega C} = \frac{1}{\mathrm{j}(1000)(20\times 10^{-6})} = -\mathrm{j}50\ \Omega, \qquad \mathrm{j}\omega L = \mathrm{j}(1000)(0.05) = \mathrm{j}50\ \Omega$$

Three elements sit in parallel between the middle node and the bottom rail: $-$j50, 100, and j50. Use admittances, which is always easier for three or more parallel branches:

$$\mathbf{Y} = \frac{1}{-\mathrm{j}50} + \frac{1}{100} + \frac{1}{\mathrm{j}50} = \mathrm{j}0.02 + 0.01 - \mathrm{j}0.02 = 0.01\ \mathrm{S}$$

The j terms cancel exactly — this network is at resonance. So $\mathbf{Z}_p = 100\,\Omega$ (purely real) and

$$\boxed{\mathbf{Z}_{ab} = 50 + 100 = 150\,\Omega \ \ \text{(purely resistive)}}$$

Watch for this. When a problem gives you an L and a C in parallel and a suspiciously convenient ω, check for cancellation before grinding through complex division. Chapter 14 formalizes this as parallel resonance.

4 Chapter 10 — Sinusoidal Steady-State Analysis

Everything from Chapters 3 and 4 carries over unchanged. The only differences are that resistances become impedances and the arithmetic is complex. Choose your method the same way you did for DC:

Circuit feature	Preferred method
Few nodes, many loops, current sources present	Nodal
Few meshes, planar, voltage sources present	Mesh
Two or more sources at *different* frequencies	Superposition (mandatory)
One branch of interest, rest of circuit fixed	Thévenin / Norton
Series voltage source ↔ parallel current source	Source transformation

Set up the matrix, then let the calculator do the algebra. Cramer's rule on a 2×2 complex system is fast by hand:

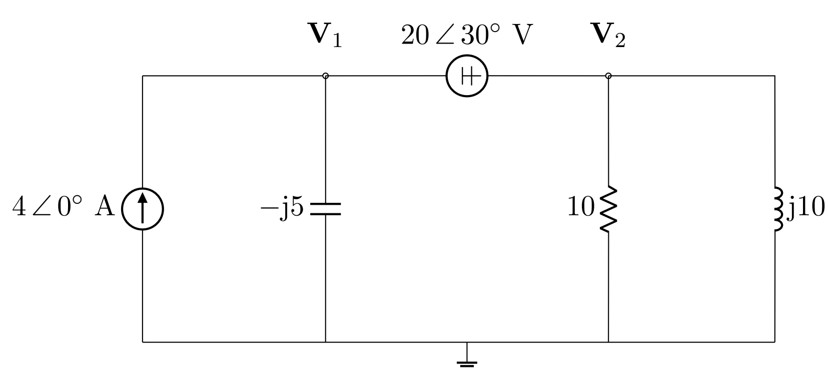
$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} \Rightarrow \Delta = ad - bc, \quad x_1 = \frac{e_1 d - b e_2}{\Delta}, \quad x_2 = \frac{a e_2 - e_1 c}{\Delta}$$

Do all multiplication and division in polar form and all addition and subtraction in rectangular form. Converting back and forth is where errors creep in, so write both forms next to each other for every intermediate result.

4.1 Worked problem 10.1 — Nodal analysis with a supernode

Problem.

Find $\mathbf{V}_1$ and $\mathbf{V}_2$ in the circuit below.



Solution. A voltage source with no ground reference sits between nodes 1 and 2, so enclose both in a supernode. Two equations are needed: KCL on the supernode plus the constraint from the source.

Constraint (the easy one — write it first):

$$\mathbf{V}_1 - \mathbf{V}_2 = 20\angle 30^\circ \quad\Rightarrow\quad \mathbf{V}_1 = \mathbf{V}_2 + 20\angle 30^\circ$$

KCL on the supernode (current in = current out; every branch leaving the supernode to ground counts):

$$4\angle 0^\circ = \frac{\mathbf{V}_1}{-\mathrm{j}5} + \frac{\mathbf{V}_2}{10} + \frac{\mathbf{V}_2}{\mathrm{j}10}$$

Substitute the constraint:

$$4 = \frac{\mathbf{V}_2 + 20\angle 30^\circ}{-\mathrm{j}5} + \mathbf{V}_2\,(0.1 - \mathrm{j}0.1)$$
$$4 = \mathbf{V}_2(\mathrm{j}0.2 + 0.1 - \mathrm{j}0.1) + 20\angle 30^\circ \cdot \mathrm{j}0.2$$
$$4 = \mathbf{V}_2(0.1 + \mathrm{j}0.1) + 4\angle 120^\circ$$

Since $4\angle 120^\circ = -2 + \mathrm{j}3.464$,

$$6 - \mathrm{j}3.464 = \mathbf{V}_2(0.1 + \mathrm{j}0.1) \quad\Rightarrow\quad \mathbf{V}_2 = \frac{6.928\angle -30^\circ}{0.1414\angle 45^\circ} = 48.99\angle -75^\circ\ \mathrm{V}$$

$$\mathbf{V}_1 = 48.99\angle -75^\circ + 20\angle 30^\circ = (12.68 - \mathrm{j}47.32) + (17.32 + \mathrm{j}10) = 30 - \mathrm{j}37.32$$

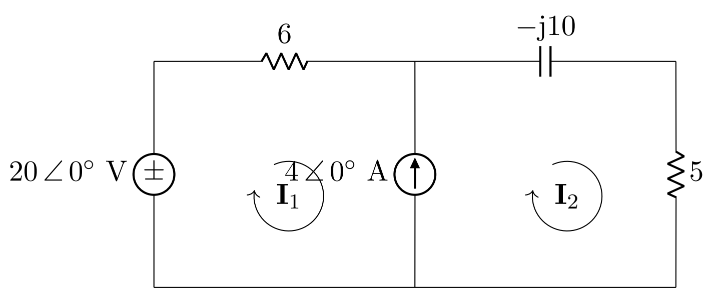
$$\boxed{\mathbf{V}_1 = 47.88\angle -51.21^\circ\ \mathrm{V}, \qquad \mathbf{V}_2 = 48.99\angle -75^\circ\ \mathrm{V}}$$

Check. Plug back into KCL: $\mathbf{V}_1/(-\mathrm{j}5) = 9.577\angle 38.79^\circ = 7.464 + \mathrm{j}6.0$; $\mathbf{V}_2/10 = 1.268 - \mathrm{j}4.732$; $\mathbf{V}_2/(\mathrm{j}10) = 4.899\angle -165^\circ = -4.732 - \mathrm{j}1.268$. Sum $= 4.0 + \mathrm{j}0.0$. ✓

4.2 Worked problem 10.2 — Mesh analysis with a supermesh

Problem.

Use mesh analysis to find $\mathbf{I}_1$ and $\mathbf{I}_2$.



Solution. The 4 A source is in the branch shared by both meshes, so it fixes the difference of the mesh currents and its own voltage is unknown. Form a supermesh by removing that branch.

Constraint (from the current source, with $\mathbf{I}_1$ *and* $\mathbf{I}_2$ *both clockwise and the source arrow pointing down through the shared branch):*

$$\mathbf{I}_2 - \mathbf{I}_1 = 4\angle 0^\circ$$

KVL around the supermesh (the outer loop, skipping the shared branch):

$$-20\angle 0^\circ + 6\mathbf{I}_1 + (-\mathrm{j}10)\mathbf{I}_2 + 5\mathbf{I}_2 = 0$$

$$6\mathbf{I}_1 + (5 - \mathrm{j}10)\mathbf{I}_2 = 20$$

Substitute $\mathbf{I}_2 = \mathbf{I}_1 + 4$:

$$6\mathbf{I}_1 + (5 - \mathrm{j}10)(\mathbf{I}_1 + 4) = 20 \quad \Rightarrow \quad (11 - \mathrm{j}10)\mathbf{I}_1 = 20 - 20 + \mathrm{j}40 = \mathrm{j}40$$

$$\mathbf{I}_1 = \frac{40\angle 90^\circ}{14.866\angle -42.27^\circ} = 2.691\angle 132.27^\circ \text{ A}$$

$$\mathbf{I}_2 = \mathbf{I}_1 + 4 = (-1.81 + \mathrm{j}1.991) + 4 = 2.19 + \mathrm{j}1.991 = 2.960\angle 42.27^\circ \text{ A}$$

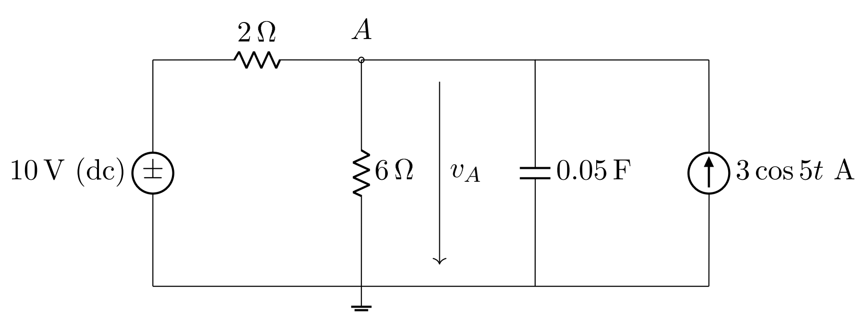
$$\boxed{\mathbf{I}_1 = 2.691\angle 132.27^\circ \text{ A}, \qquad \mathbf{I}_2 = 2.960\angle 42.27^\circ \text{ A}}$$

Supermesh checklist. (1) The current source is in a shared branch ⇒ supermesh. (2) The constraint equation always comes from the source, and its sign depends on the source's arrow direction relative to your assumed mesh currents — draw the arrow on the schematic and read it off, do not guess. (3) The KVL loop goes around the outside and never crosses the removed branch.

4.3 Worked problem 10.3 — Superposition with two frequencies

Problem.

Find $v_A(t)$ in the circuit below. Note the two sources have different frequencies.



Solution. Superposition is not optional here. Two frequencies cannot coexist in one phasor diagram, because $\mathrm{j}\omega L$ and $1/\mathrm{j}\omega C$ would need two values at once. Solve each frequency in its own circuit and add the *time-domain* results.

Case 1: DC source alone ($\omega = 0$), current source opened.

At DC a capacitor is an open circuit and an inductor is a short. The capacitor branch carries nothing, so the circuit is a simple divider:

$$v_{A1} = 10 \cdot \frac{6}{2+6} = 7.5\,\mathrm{V}$$

Case 2: AC source alone ($\omega = 5$ rad/s), DC voltage source shorted.

$$\frac{1}{\mathrm{j}\omega C} = \frac{1}{\mathrm{j}(5)(0.05)} = -\mathrm{j}4\ \Omega$$

With the 10 V source replaced by a short, the 2 Ω resistor now runs from node A straight to ground. So three branches hang off node A: 2, 6, and −j4, all in parallel, driven by $3\angle 0^\circ$ A.

$$\mathbf{Y} = \frac{1}{2} + \frac{1}{6} + \frac{1}{-\mathrm{j}4} = 0.5 + 0.1667 + \mathrm{j}0.25 = 0.6667 + \mathrm{j}0.25 = 0.7120\angle 20.56^\circ\ \mathrm{S}$$

$$\mathbf{V}_{A2} = \frac{3\angle 0^\circ}{0.7120\angle 20.56^\circ} = 4.214\angle -20.56^\circ\ \mathrm{V} \ \Rightarrow\ v_{A2}(t) = 4.214\cos(5t - 20.56^\circ)\ \mathrm{V}$$

Add in the time domain:

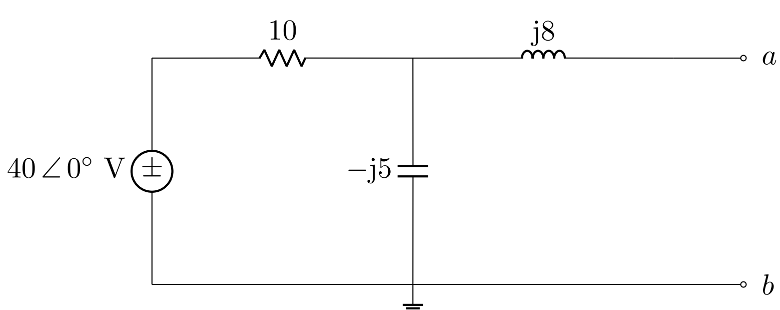
$$\boxed{v_A(t) = 7.5 + 4.214\cos(5t - 20.56^\circ)\ \mathrm{V}}$$

The trap. You cannot write this as a single phasor. If an exam problem asks for a phasor answer and the sources differ in frequency, the question is malformed or you have misread it. Also note that "turn off a voltage source" means *short* it and "turn off a current source" means *open* it — the resistor topology changes between the two cases, which is exactly why the 2 Ω resistor participates in Case 2 but acts as part of a divider in Case 1.

4.4 Worked problem 10.4 — Thévenin and Norton equivalents

Problem.

Find the Thévenin and Norton equivalents at terminals a–b.



Solution.

Step 1 — $\mathbf{V}_{Th}$ (open-circuit voltage). With *a–b* open, no current flows through the j8 inductor, so it drops zero volts and $\mathbf{V}_{\mathrm{Th}}$ equals the voltage across the capacitor. Voltage divider:

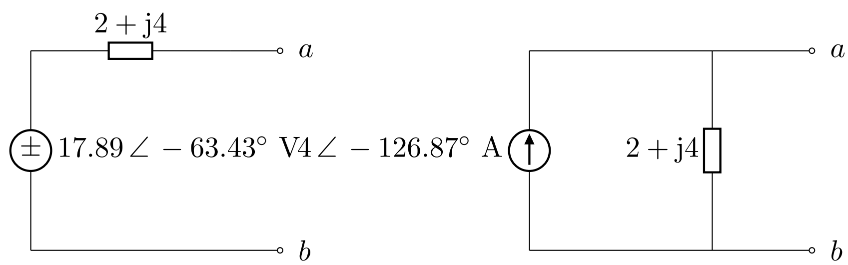
$$\mathbf{V}_{\mathrm{Th}} = 40\angle 0° \cdot \frac{-\mathrm{j}5}{10-\mathrm{j}5} = \frac{200\angle -90°}{11.18\angle -26.57°} = 17.89\angle -63.43° \text{ V}$$

Step 2 — $\mathbf{Z}_{Th}$ (kill the independent source). Short the voltage source. Then the $10\,\Omega$ and $-$j5 are in parallel, and that combination is in series with j8:

$$\mathbf{Z}_{\mathrm{Th}} = \mathrm{j}8 + \frac{(10)(-\mathrm{j}5)}{10-\mathrm{j}5} = \mathrm{j}8 + (2-\mathrm{j}4) = 2+\mathrm{j}4\ \Omega = 4.472\angle 63.43°\ \Omega$$

Step 3 — Norton.

$$\mathbf{I}_N = \frac{\mathbf{V}_{\mathrm{Th}}}{\mathbf{Z}_{\mathrm{Th}}} = \frac{17.89\angle -63.43°}{4.472\angle 63.43°} = 4\angle -126.87° \text{ A}, \qquad \mathbf{Z}_N = \mathbf{Z}_{\mathrm{Th}} = 2+\mathrm{j}4\ \Omega$$



Thévenin

Norton

> **If the circuit contains a dependent source**, you cannot simply zero sources and combine impedances. Use one of:
>
> - $\mathbf{Z}_{\mathrm{Th}} = \mathbf{V}_{\mathrm{Th}}/\mathbf{I}_{sc}$ — find open-circuit voltage and short-circuit current separately, or
> - apply a $1\angle 0°$ V test source at *a–b* with all *independent* sources killed, and compute $\mathbf{Z}_{\mathrm{Th}} = 1/\mathbf{I}_{\mathrm{test}}$.

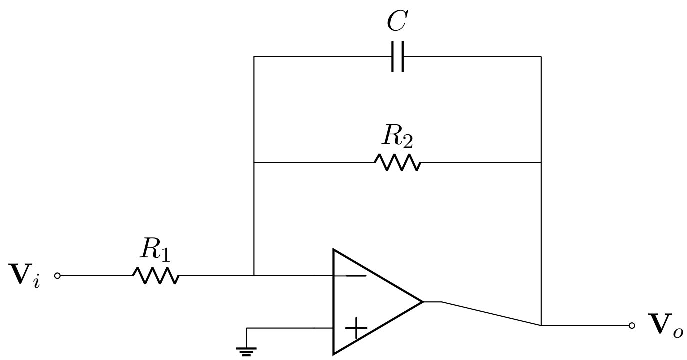
4.5 Op amp AC circuits

Ideal op amp rules survive intact in the phasor domain: $\mathbf{V}_+ = \mathbf{V}_-$ and no current into either input. The standard gains just take impedances:

$$\text{inverting: } \mathbf{H} = -\frac{\mathbf{Z}_f}{\mathbf{Z}_i} \qquad \text{non-inverting: } \mathbf{H} = 1 + \frac{\mathbf{Z}_f}{\mathbf{Z}_i}$$

Quick example.

For the inverting configuration with $\mathbf{Z}_i = R_1 = 10\,\mathrm{k}\Omega$ and $\mathbf{Z}_f$ equal to $R_2 = 50\,\mathrm{k}\Omega$ in parallel with $C = 2\,\mathrm{nF}$, find the gain at $\omega = 1\times 10^4\,\mathrm{rad\,s^{-1}}$.



$\mathbf{Z}_f = R_2 \parallel \dfrac{1}{\mathrm{j}\omega C} = \dfrac{R_2}{1+\mathrm{j}\omega R_2 C}$, and $\omega R_2 C = (10^4)(5\times10^4)(2\times10^{-9}) = 1$, so $\mathbf{Z}_f = \dfrac{50\text{k}}{1+\mathrm{j}1} = 35.36\angle-45^\circ$ kΩ.

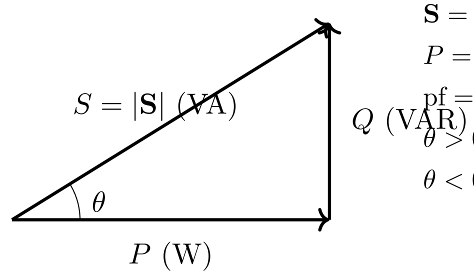
$$\mathbf{H} = -\frac{35.36\angle-45^\circ}{10} = 3.536\angle 135^\circ$$

The magnitude is 3.54 and the phase is 135° (the 180° from the inversion minus the 45° lag from the feedback capacitor).

5 Chapter 11 — AC Power Analysis

Before you compute anything, decide whether the phasors are amplitude or RMS. Sadiku writes $\mathbf{S} = \frac{1}{2}\mathbf{V}\mathbf{I}^*$ for amplitude phasors and $\mathbf{S} = \mathbf{V}_{\text{rms}}\mathbf{I}^*_{\text{rms}}$ for RMS phasors. A factor of two in your final power is almost always this. Rule of thumb: if the problem gives you a time function like $10\cos(\omega t)$, you have amplitudes and you need the $\frac{1}{2}$. If the problem says "120 V" for a wall outlet or gives a three-phase system, you have RMS and there is no $\frac{1}{2}$.

5.1 The power triangle



$\mathbf{S} = P + \mathrm{j}Q$

$P = S\cos\theta,\ \ Q = S\sin\theta$

$\mathrm{pf} = \cos\theta = P/S$

$\theta > 0$: inductive, lagging pf

$\theta < 0$: capacitive, leading pf

θ is simultaneously the angle of $\mathbf{Z}$, the angle of $\mathbf{S}$, and the amount by which voltage leads current. Those three statements are the same statement.

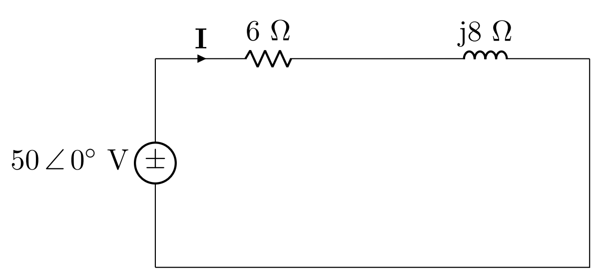
Conservation. Complex power is conserved: $\sum \mathbf{S}_{\text{supplied}} = \sum \mathbf{S}_{\text{absorbed}}$, and therefore P and Q are each conserved separately. Apparent power S is *not* conserved — you cannot add magnitudes. Add loads as $P + \mathrm{j}Q$, then take the magnitude at the end.

5.2 Worked problem 11.1 — Average power and the power triangle

---

Problem.

A source $v_s(t) = 50\cos(\omega t)$ V drives a series RL branch with $R = 6\,\Omega$ and $\mathrm{j}\omega L = \mathrm{j}8\,\Omega$. Find (a) the average power absorbed by each element, (b) the complex power delivered by the source, and (c) the power factor.



Solution.

$$\mathbf{Z} = 6 + \mathrm{j}8 = 10\angle 53.13^\circ\ \Omega \qquad \mathbf{I} = \frac{50\angle 0^\circ}{10\angle 53.13^\circ} = 5\angle -53.13^\circ\ \text{A}$$

(a) *Resistor:* $P_R = \frac{1}{2} I_m^2 R = \frac{1}{2}(5)^2(6) = 75\,\text{W}$.
Inductor: $P_L = 0$. An inductor absorbs zero average power over a full cycle — it stores energy for a quarter cycle and returns it the next. Same for a capacitor.

(b) $\mathbf{S} = \frac{1}{2}\mathbf{V}\mathbf{I}^* = \frac{1}{2}(50\angle 0°)(5\angle +53.13°) = 125\angle 53.13° = 75 + \text{j}100$ VA.

So the source delivers 75 W of real power (all of it consumed by the resistor, as it must be) and 100 VAR of reactive power (all of it exchanged with the inductor). The apparent power is 125 VA.

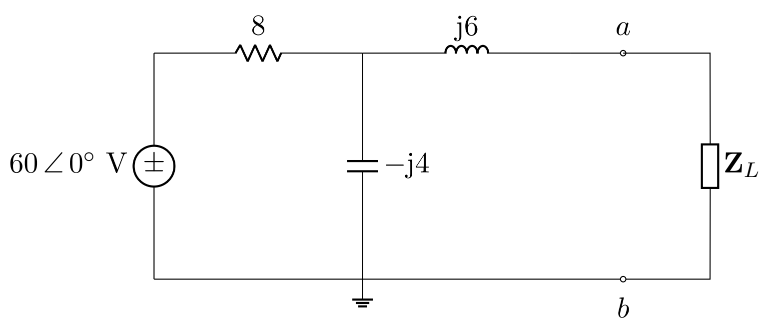
(c) pf $= \cos 53.13° = 0.6$ **lagging**, because the load is inductive and the current lags the voltage.

RMS cross-check. $V_{\text{rms}} = 50/\sqrt{2} = 35.36\,\text{V}$, $I_{\text{rms}} = 5/\sqrt{2} = 3.536\,\text{A}$, so $S = (35.36)(3.536) = 125\,\text{VA}$ and $P = 125(0.6) = 75\,\text{W}$. ✓

5.3 Worked problem 11.2 — Maximum average power transfer

Problem.

Find the load impedance $\mathbf{Z}_L$ that absorbs maximum average power, and compute that power.



Solution. Remove $\mathbf{Z}_L$ and find the Thévenin equivalent at *a*–*b*.

$\mathbf{Z}_{Th}$*:* short the source; then 8 ∥ (−j4) in series with j6:

$$\frac{(8)(-\text{j}4)}{8-\text{j}4} = \frac{32\angle -90°}{8.944\angle -26.57°} = 3.578\angle -63.43° = 1.6 - \text{j}3.2$$

$$\mathbf{Z}_{\text{Th}} = \text{j}6 + (1.6 - \text{j}3.2) = 1.6 + \text{j}2.8\ \Omega$$

$\mathbf{V}_{Th}$*:* with *a*–*b* open, no current in j6, so $\mathbf{V}_{\text{Th}}$ is the capacitor voltage:

$$\mathbf{V}_{\text{Th}} = 60\angle 0° \cdot \frac{-\text{j}4}{8-\text{j}4} = 60\angle 0° \cdot 0.4472\angle -63.43° = 26.83\angle -63.43°\ \text{V}$$

Matched load:

$$\boxed{\mathbf{Z}_L = \mathbf{Z}_{\text{Th}}^* = 1.6 - \text{j}2.8\ \Omega}$$

This is a $1.6\,\Omega$ resistor in series with a capacitive reactance of $2.8\,\Omega$. The reactances cancel, leaving $\mathbf{Z}_{\text{total}} = 3.2\ \Omega$ purely resistive.

Maximum power (amplitude phasors, so use the /8 form):

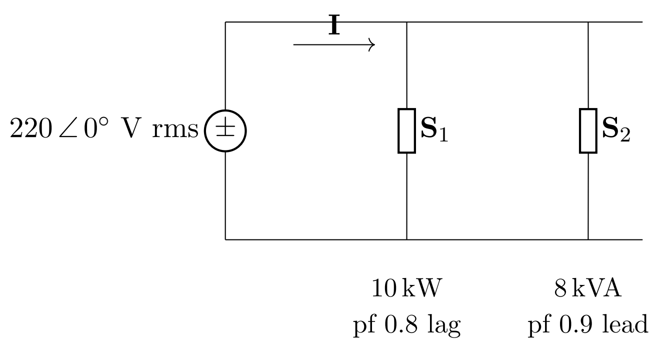
$$P_{\max} = \frac{|\mathbf{V}_{\text{Th}}|^2}{8R_{\text{Th}}} = \frac{(26.83)^2}{8(1.6)} = \frac{720}{12.8} = \boxed{56.25\,\text{W}}$$

Variation to expect. If the problem restricts $\mathbf{Z}_L$ to a pure resistance R_L, the answer changes: $R_L = |\mathbf{Z}_{\text{Th}}| = \sqrt{1.6^2 + 2.8^2} = 3.224\,\Omega$, and the power is then lower than 56.25 W because the reactances no longer cancel.

5.4 Worked problem 11.3 — Conservation of AC power

Problem.

A 220 V rms, 60 Hz source feeds two loads in parallel. Load 1 absorbs 10 kW at pf 0.8 lagging. Load 2 absorbs 8 kVA at pf 0.9 leading. Find the total complex power, the overall power factor, and the source current.



Solution. Convert each load to rectangular complex power, then add.

Load 1: $\theta_1 = \cos^{-1}(0.8) = +36.87^\circ$ (positive, lagging). $P_1 = 10\,\text{kW}$, so $S_1 = P_1/\text{pf} = 10/0.8 = 12.5\,\text{kVA}$ and

$$\mathbf{S}_1 = 12.5\angle 36.87^\circ = 10 + \text{j}7.5 \text{ kVA}$$

Load 2: $\theta_2 = -\cos^{-1}(0.9) = -25.84^\circ$ (negative, leading). $S_2 = 8\,\text{kVA}$, so

$$\mathbf{S}_2 = 8\angle -25.84^\circ = 7.2 - \text{j}3.487 \text{ kVA}$$

Total:

$$\mathbf{S}_T = \mathbf{S}_1 + \mathbf{S}_2 = 17.2 + \text{j}4.013 \text{ kVA} = 17.66\angle 13.13^\circ \text{ kVA}$$

$$\boxed{P_T = 17.2\,\text{kW}, \quad Q_T = 4.013\,\text{kVAR}, \quad \text{pf} = \cos 13.13^\circ = 0.974 \text{ lagging}}$$

Source current: using $\mathbf{S} = \mathbf{V}_{\text{rms}}\mathbf{I}^*_{\text{rms}}$,

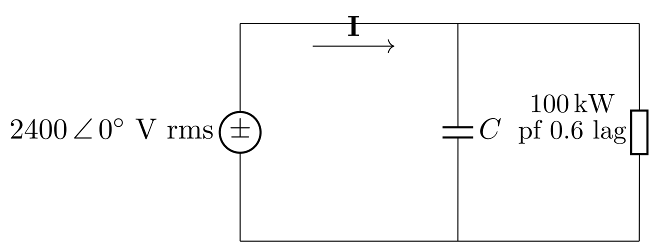
$$\mathbf{I}^*_{\text{rms}} = \frac{17660\angle 13.13^\circ}{220\angle 0^\circ} = 80.28\angle 13.13^\circ \quad \Rightarrow \quad \mathbf{I}_{\text{rms}} = 80.28\angle -13.13^\circ \text{ A}$$

Note what did not work. $S_1 + S_2 = 12.5 + 8 = 20.5\,\text{kVA}$, but the true apparent power is 17.66 kVA. Apparent powers do not add. The leading VARs of load 2 partially cancel the lagging VARs of load 1, which is exactly the mechanism behind power factor correction.

5.5 Worked problem 11.4 — Power factor correction

Problem.

A 100 kW industrial load operates at pf 0.6 lagging from a 2400 V rms, 60 Hz line. Find the parallel capacitance needed to raise the power factor to 0.95 lagging, and the reduction in line current.



Solution. A parallel capacitor changes Q without changing P. That is the whole idea: the real power delivered to the load is untouched, but the reactive power the source must supply drops.

Before: $\theta_1 = \cos^{-1}(0.6) = 53.13°$

$$Q_1 = P\tan\theta_1 = 100\tan(53.13°) = 133.33\,\text{kVAR}$$

After: $\theta_2 = \cos^{-1}(0.95) = 18.19°$

$$Q_2 = P\tan\theta_2 = 100\tan(18.19°) = 32.87\,\text{kVAR}$$

Capacitor must supply the difference:

$$Q_C = Q_1 - Q_2 = 100.46\,\text{kVAR}$$

$$C = \frac{Q_C}{\omega V_{\text{rms}}^2} = \frac{100{,}460}{2\pi(60)(2400)^2} = \frac{100{,}460}{2.171\times10^9} = \boxed{46.3\,\mu\text{F}}$$

Line current before and after:

$$I_1 = \frac{S_1}{V} = \frac{100{,}000/0.6}{2400} = 69.4\,\text{A} \qquad I_2 = \frac{100{,}000/0.95}{2400} = 43.9\,\text{A}$$

A 37 % reduction in line current for the same delivered real power. Since line losses go as I^2, they drop by about 60 %. This is why utilities penalize low power factor.

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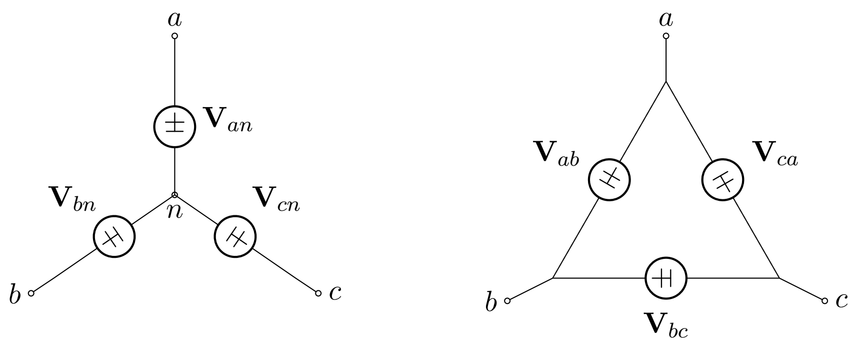
6 Chapter 12 — Three-Phase Circuits

6.1 Get the geometry right first

Almost every three-phase error is a $\sqrt{3}$ or a 30° in the wrong place. Fix the two pictures below in your head and the rest is bookkeeping.



Y (wye) source   Δ (delta) source

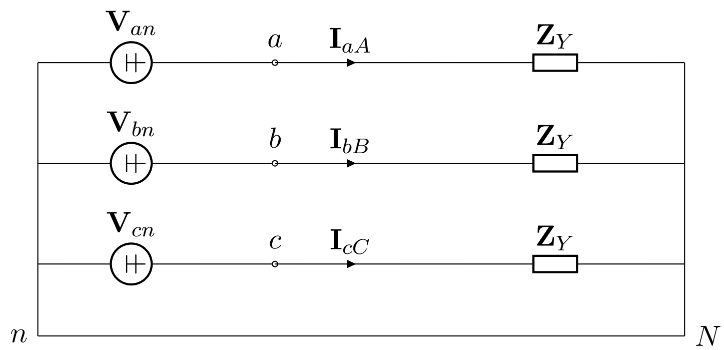
Y connection	Δ connection
Line current = phase current: $\mathbf{I}_a = \mathbf{I}_{aN}$	Line voltage = phase voltage: $\mathbf{V}_{ab} = \mathbf{V}_{AB}$
Line voltage is $\sqrt{3}$ larger and 30° ahead: $\mathbf{V}_{ab} = \sqrt{3}\,\mathbf{V}_{an}\angle{+30°}$	Line current is $\sqrt{3}$ larger and 30° behind: $\mathbf{I}_a = \sqrt{3}\,\mathbf{I}_{AB}\angle{-30°}$

The universal method. Convert any Δ load to Y using $\mathbf{Z}_Y = \mathbf{Z}_\Delta/3$, convert a Δ source to Y using $V_p = V_L/\sqrt{3}$ with a −30° shift, then analyze *one phase* of a Y–Y circuit as an ordinary single-loop AC problem. Get the other two phases by subtracting and adding 120°. In a balanced system the neutral carries no current, so you can tie n and N together whether or not a neutral wire exists.

6.2 Worked problem 12.1 — Balanced Y–Y system

Problem.

A balanced *abc*-sequence Y-connected source with $\mathbf{V}_{an} = 220\angle 0°$ V rms supplies a balanced Y-connected load with $\mathbf{Z}_Y = 12 + j9\ \Omega$ per phase. Line impedance is negligible. Find the line currents, the line voltage, and the total real, reactive, and apparent power.



Solution. Balanced, so analyze phase a alone.

$$\mathbf{Z}_Y = 12 + \mathrm{j}9 = 15\angle 36.87^\circ\ \Omega$$

$$\mathbf{I}_{aA} = \frac{\mathbf{V}_{an}}{\mathbf{Z}_Y} = \frac{220\angle 0^\circ}{15\angle 36.87^\circ} = 14.67\angle -36.87^\circ\ \mathrm{A}$$

By symmetry (abc sequence, so subtract 120° going $a \to b \to c$):

$$\boxed{\mathbf{I}_{aA} = 14.67\angle -36.87^\circ\ \mathrm{A}, \quad \mathbf{I}_{bB} = 14.67\angle -156.87^\circ\ \mathrm{A}, \quad \mathbf{I}_{cC} = 14.67\angle 83.13^\circ\ \mathrm{A}}$$

Note $\mathbf{I}_{aA} + \mathbf{I}_{bB} + \mathbf{I}_{cC} = 0$, which is why the neutral wire carries no current.

Line voltage: $V_L = \sqrt{3}(220) = 381.1\,\mathrm{V}$ rms, with $\mathbf{V}_{ab} = 381.1\angle 30^\circ$ V.

Power. For a Y load, $I_L = I_p = 14.67$ A and $\theta = 36.87^\circ$:

$$S_T = \sqrt{3}\,V_L I_L = \sqrt{3}(381.1)(14.67) = 9680\,\mathrm{VA}$$

$$P_T = S_T\cos\theta = 9680(0.8) = 7744\,\mathrm{W} \qquad Q_T = S_T\sin\theta = 9680(0.6) = 5808\,\mathrm{VAR}$$

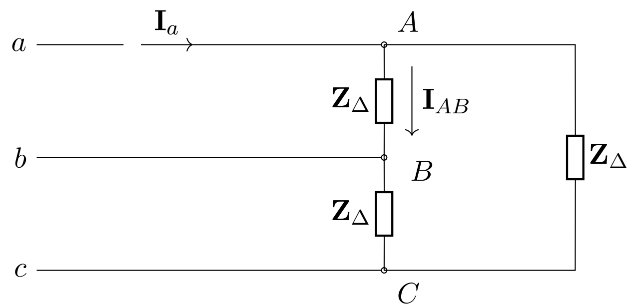
Cross-check with the per-phase form: $P_T = 3I_p^2R = 3(14.67)^2(12) = 7744\,\mathrm{W}$. ✓

6.3 Worked problem 12.2 — Balanced Δ load and the two-wattmeter method

---

Problem.

A balanced Δ-connected load of $\mathbf{Z}_\Delta = 30 + \mathrm{j}40\ \Omega$ per phase is supplied by a balanced abc source with $\mathbf{V}_{ab} = 440\angle 0^\circ$ V rms. Find (a) the phase and line currents, (b) the total power, and (c) the readings of two wattmeters connected in lines a and b with their voltage coils referenced to line c.



Δ-connected load

Solution.

(a) For a Δ load the phase voltage *is* the line voltage:

$$\mathbf{Z}_\Delta = 30 + \mathrm{j}40 = 50\angle 53.13^\circ\ \Omega \qquad \mathbf{I}_{AB} = \frac{440\angle 0^\circ}{50\angle 53.13^\circ} = 8.8\angle -53.13^\circ\ \mathrm{A}$$

Line current is $\sqrt{3}$ larger and lags the phase current by 30°:

$$\mathbf{I}_a = \sqrt{3}(8.8)\angle(-53.13^\circ - 30^\circ) = 15.24\angle -83.13^\circ\ \mathrm{A}$$

$$\boxed{I_p = 8.8\,\mathrm{A}, \quad I_L = 15.24\,\mathrm{A}}$$

(b) $\theta = 53.13^\circ$ (the load impedance angle, *not* 83.13°):

$$S_T = \sqrt{3}\,V_L I_L = \sqrt{3}(440)(15.24) = 11\,616\,\mathrm{VA}$$

$$P_T = 11616\cos(53.13^\circ) = 6970\,\mathrm{W}, \qquad Q_T = 11616\sin(53.13^\circ) = 9293\,\mathrm{VAR}$$

Check: $P_T = 3I_p^2R = 3(8.8)^2(30) = 6970\,\mathrm{W}.$ ✓

(c) Two-wattmeter method:

$$P_1 = V_L I_L\cos(\theta + 30^\circ) = (440)(15.24)\cos(83.13^\circ) = 802\,\mathrm{W}$$

$$P_2 = V_L I_L\cos(\theta - 30^\circ) = (440)(15.24)\cos(23.13^\circ) = 6167\,\mathrm{W}$$

$$P_1 + P_2 = 6969\,\mathrm{W} = P_T \checkmark \qquad \sqrt{3}(P_2 - P_1) = \sqrt{3}(5365) = 9292\,\mathrm{VAR} = Q_T \checkmark$$

Two-wattmeter facts worth memorizing.

- Two wattmeters are always sufficient for a three-wire system, balanced or not.
- If $\theta = 0$ (unity pf), the meters read equally.
- If $\theta = 60^\circ$, one meter reads zero.
- If $\theta > 60^\circ$, one meter reads *negative* — the pointer tries to go backwards and you must reverse a coil and record the reading as negative.
- $P_2 > P_1$ for a lagging load; the meter in the lagging line reads less.

6.4 Unbalanced systems

If the load is unbalanced you lose per-phase analysis and must fall back on ordinary mesh or nodal analysis of the full three-phase circuit.

- **Unbalanced four-wire Y–Y:** each phase is independent because the neutral holds $V_{nN} = 0$. Compute $\mathbf{I}_{aA} = \mathbf{V}_{an}/\mathbf{Z}_A$ and so on, then the neutral current is $\mathbf{I}_{nN} = -(\mathbf{I}_{aA} + \mathbf{I}_{bB} + \mathbf{I}_{cC})$, which is nonzero.
- **Unbalanced three-wire Y–Y:** the load neutral floats. Use mesh analysis with two loop currents.
- **Total power** is the sum of the three individual phase powers; the $\sqrt{3}$ formulas do not apply.

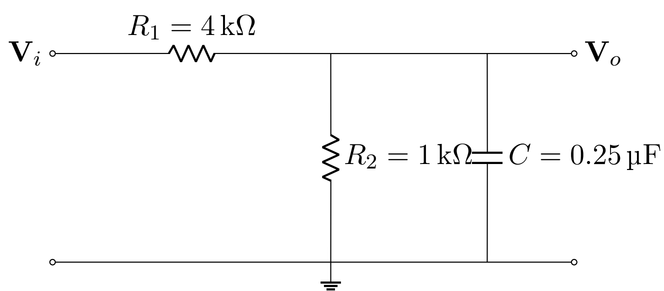
7 Chapter 14 — Frequency Response

The shift here is that ω becomes the variable instead of a fixed number. Everything from Chapters 9 and 10 still applies; you just carry ω symbolically through the algebra instead of substituting a value at step one.

7.1 Worked problem 14.1 — Transfer function from a schematic

Problem.

Find $\mathbf{H}(\omega) = \mathbf{V}_o/\mathbf{V}_i$ for the circuit below. Identify the filter type, the DC gain, and the corner frequency.



Solution. The output is taken across the parallel combination of R_2 and C, so this is a voltage divider with R_1 on top.

$$\mathbf{Z}_2 = R_2 \parallel \frac{1}{\mathrm{j}\omega C} = \frac{R_2}{1 + \mathrm{j}\omega R_2 C}$$

$$\mathbf{H}(\omega) = \frac{\mathbf{Z}_2}{R_1 + \mathbf{Z}_2} = \frac{R_2}{R_1(1 + \mathrm{j}\omega R_2 C) + R_2} = \frac{R_2}{(R_1 + R_2) + \mathrm{j}\omega R_1 R_2 C}$$

Substitute: $R_1 R_2 C = (4000)(1000)(0.25 \times 10^{-6}) = 1$, and $R_1 + R_2 = 5000$:

$$\boxed{\mathbf{H}(\omega) = \frac{1000}{5000 + \mathrm{j}\omega} = \frac{0.2}{1 + \mathrm{j}\omega/5000}}$$

Reading the answer. The standard form $K/(1 + \mathrm{j}\omega/\omega_c)$ is a **low-pass** filter with

$$H(0) = 0.2 \ (= -13.98\,\mathrm{dB}), \qquad \omega_c = 5000\,\mathrm{rad\,s^{-1}}, \qquad f_c = \frac{5000}{2\pi} = 795.8\,\mathrm{Hz}$$

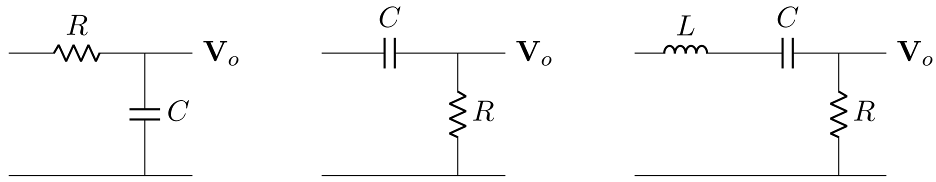
At ω_c the magnitude is $0.2/\sqrt{2} = 0.1414$, down 3 dB from DC, and the phase is $-45°$.

Two sanity checks you should always run on a transfer function. As $\omega \to 0$, C is an open circuit, so the divider is $R_2/(R_1 + R_2) = 0.2$. ✓As $\omega \to \infty$, C is a short, so $\mathbf{V}_o \to 0$. ✓Those two limits alone tell you the filter type without any algebra.

7.2 Identifying passive filters by inspection

Replace C and L with their DC and high-frequency limits and see what happens to $\mathbf{V}_o$:

	$\omega \to 0$	$\omega \to \infty$
Capacitor	open	short
Inductor	short	open



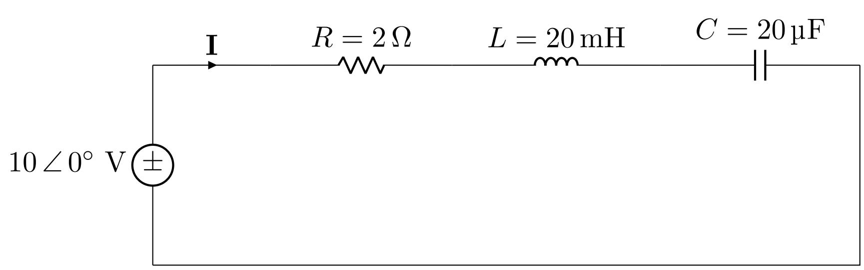
Low-pass $\omega_c = 1/RC$ High-pass $\omega_c = 1/RC$ Band-pass ($\mathbf{V}_o$ across R)

A series RLC with the output across R is band-pass; with the output across the LC pair it is band-stop (a notch). Both have $\omega_0 = 1/\sqrt{LC}$ and $B = R/L$.

7.3 Worked problem 14.2 — Series resonance

Problem.

A series RLC circuit has $R = 2\,\Omega$, $L = 20\,\text{mH}$, $C = 20\,\mu\text{F}$, driven by $v_s(t) = 10\cos\omega t$ V. Find ω_0, Q, B, the half-power frequencies, the current at resonance, and the inductor voltage at resonance.



Solution.

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(0.02)(20\times10^{-6})}} = \frac{1}{\sqrt{4\times10^{-7}}} = 1581.1\,\text{rad}\,\text{s}^{-1} \quad (f_0 = 251.6\,\text{Hz})$$

$$Q = \frac{\omega_0 L}{R} = \frac{(1581.1)(0.02)}{2} = 15.81 \qquad B = \frac{R}{L} = \frac{2}{0.02} = 100\,\text{rad}\,\text{s}^{-1}$$

Check: $B = \omega_0/Q = 1581.1/15.81 = 100$. ✓

Since $Q = 15.8 \geq 10$ this is a high-Q circuit, so the half-power frequencies are symmetric to good accuracy:

$$\omega_{1,2} \approx \omega_0 \mp \frac{B}{2} = 1581.1 \mp 50 \ \Rightarrow\ \omega_1 \approx 1531\,\mathrm{rad\,s^{-1}},\quad \omega_2 \approx 1631\,\mathrm{rad\,s^{-1}}$$

The exact values from $\omega_{1,2} = \mp\frac{R}{2L} + \sqrt{(\frac{R}{2L})^2 + \frac{1}{LC}}$ are 1531.9 and $1631.9\,\mathrm{rad\,s^{-1}}$, confirming the approximation is good to about 0.06 %.

At resonance the reactances cancel and $\mathbf{Z} = R$:

$$\mathbf{I} = \frac{10\angle 0^\circ}{2} = 5\angle 0^\circ \text{ A}$$

This is the maximum current the circuit ever draws, and it is in phase with the source (unity power factor).

$$|\mathbf{V}_L| = |\mathbf{I}|\,\omega_0 L = (5)(1581.1)(0.02) = 158.1\,\mathrm{V} = Q \cdot V_s$$

$$\boxed{|\mathbf{V}_L| = |\mathbf{V}_C| = QV_s = 15.81 \times 10 = 158.1\,\mathrm{V}}$$

The counterintuitive part. The inductor and capacitor each sit at 158 V while the source is only 10 V. There is no contradiction: $\mathbf{V}_L$ and $\mathbf{V}_C$ are 180° out of phase and cancel exactly, so KVL is satisfied. Expect a conceptual exam question on this. It is also a real design hazard — capacitors in high-Q resonant circuits must be rated far above the supply voltage.

7.4 Worked problem 14.3 — Parallel resonance, ideal and practical

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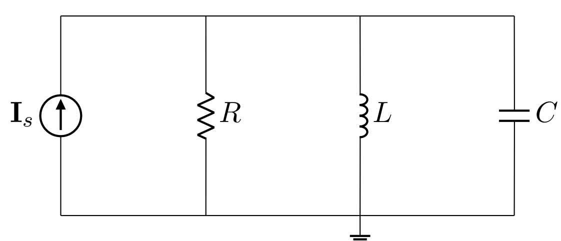
Problem (a).

An ideal parallel RLC circuit has $R = 8\,\mathrm{k\Omega}$, $L = 0.2\,\mathrm{mH}$, $C = 8\,\mathrm{nF}$. Find ω_0, Q, B.



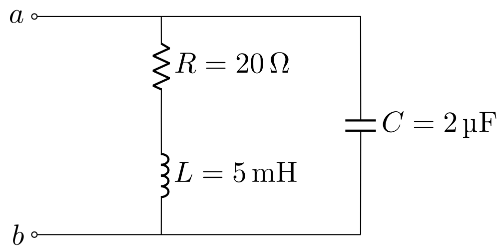
Solution.

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(2\times10^{-4})(8\times10^{-9})}} = 790.6\,\mathrm{krad\,s^{-1}}$$

$$Q = \omega_0 RC = (790{,}569)(8000)(8\times10^{-9}) = 50.6 \qquad B = \frac{1}{RC} = \frac{1}{(8000)(8\times10^{-9})} = 15.63\,\mathrm{krad\,s^{-1}}$$

$$\omega_{1,2} \approx \omega_0 \mp B/2 = 782.8 \text{ and } 798.4\,\mathrm{krad\,s^{-1}}$$

Problem (b) — the practical version. A real coil has winding resistance in *series* with its inductance. For the tank circuit below with $R = 20\,\Omega$, $L = 5\,\mathrm{mH}$, $C = 2\,\mathrm{\mu F}$, find the resonant frequency.



real coil = R in series with L

Solution. You cannot use $\omega_0 = 1/\sqrt{LC}$ here. Derive the admittance and set its imaginary part to zero:

$$\mathbf{Y} = \mathrm{j}\omega C + \frac{1}{R + \mathrm{j}\omega L} = \mathrm{j}\omega C + \frac{R - \mathrm{j}\omega L}{R^2 + \omega^2 L^2}$$

$$\mathrm{Im}\{\mathbf{Y}\} = \omega C - \frac{\omega L}{R^2 + \omega^2 L^2} = 0 \quad \Rightarrow \quad R^2 + \omega^2 L^2 = \frac{L}{C}$$

$$\boxed{\omega_0 = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}}$$

$$\frac{1}{LC} = \frac{1}{(5 \times 10^{-3})(2 \times 10^{-6})} = 10^8, \qquad \frac{R^2}{L^2} = \frac{400}{2.5 \times 10^{-5}} = 1.6 \times 10^7$$

$$\omega_0 = \sqrt{10^8 - 1.6 \times 10^7} = \sqrt{8.4 \times 10^7} = 9165\,\text{rad}\,\text{s}^{-1}$$

compared to $1/\sqrt{LC} = 10\,000\,\text{rad}\,\text{s}^{-1}$ — an 8.3 % error if you had used the ideal formula. This exact setup appears repeatedly in the Ch. 14 problem set, so recognize the series-R-with-L topology on sight.

7.5 Bode plots

Procedure.

1. Put $\mathbf{H}$ into standard form: every factor must read $(1 + \mathrm{j}\omega/\omega_k)$, never $(\omega_k + \mathrm{j}\omega)$. Factor constants out front.
2. List the corner frequencies in increasing order and mark each as a pole or zero.
3. Start at low ω: the magnitude asymptote is $20 \log_{10} |K|$ plus $\pm 20N$ dB/dec if there is a $(\mathrm{j}\omega)^{\pm N}$ factor.
4. Walk left to right. At each corner, add +20 dB/dec for a zero, −20 for a pole. The running slope is cumulative.
5. Phase: each pole contributes $-90°$ spread over two decades, from $\omega_k/10$ to $10\omega_k$ (slope $-45°$/dec); each zero contributes $+90°$ the same way.

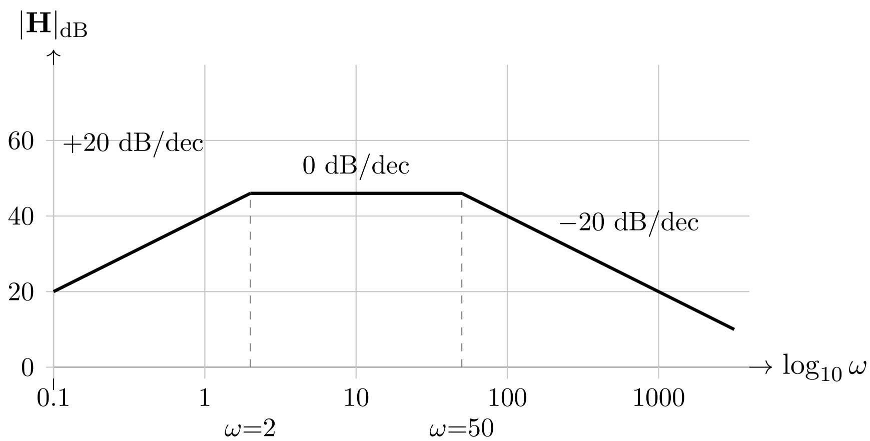
Example.

Sketch the magnitude Bode plot of $\mathbf{H}(s) = \dfrac{10^4 s}{(s+2)(s+50)}$, with $s = \mathrm{j}\omega$.

Standard form:

$$\mathbf{H} = \frac{10^4 \mathrm{j}\omega}{2\left(1+\frac{\mathrm{j}\omega}{2}\right)\cdot 50\left(1+\frac{\mathrm{j}\omega}{50}\right)} = \frac{100\,\mathrm{j}\omega}{\left(1+\frac{\mathrm{j}\omega}{2}\right)\left(1+\frac{\mathrm{j}\omega}{50}\right)}$$

Factors: constant 100 (+40 dB), a zero at the origin, a pole at $\omega = 2$, a pole at $\omega = 50$.



The flat middle band sits at $20\log_{10}(100 \times 2) = 46\,\mathrm{dB}$. This shape — rise, plateau, fall — is the signature of a band-pass response. The phase runs $+90° \to 0° \to -90°$ over the same range.

7.6 Active filters and scaling

First-order active low-pass (inverting):

$$\mathbf{H}(\omega) = -\frac{R_f}{R_i}\cdot\frac{1}{1+\mathrm{j}\omega R_f C_f}, \qquad \omega_c = \frac{1}{R_f C_f}, \qquad \text{passband gain} = -\frac{R_f}{R_i}$$

Swapping the capacitor to the input branch (in series with R_i) makes it high-pass with $\omega_c = 1/(R_i C_i)$. Cascading a low-pass stage with a high-pass stage gives a band-pass filter, which is exactly the topology in your voice-band filter lab.

Scaling. Designers work with normalized prototypes ($R = 1\,\Omega$, $\omega_0 = 1\,\mathrm{rad\,s^{-1}}$) and scale at the end.

Example.

A prototype has $R = 1\,\Omega$, $L = 1\,\mathrm{H}$, $C = 1\,\mathrm{F}$ (so $\omega_0 = 1\,\mathrm{rad\,s^{-1}}$). Scale it to run at $\omega_0 = 1 \times 10^4\,\mathrm{rad\,s^{-1}}$ in a $50\,\Omega$ system.

$K_m = 50$ (to turn $1\,\Omega$ into $50\,\Omega$) and $K_f = 10^4$:

$$R' = K_m R = 50\,\Omega \qquad L' = \frac{K_m}{K_f} L = \frac{50}{10^4} = 5\,\text{mH} \qquad C' = \frac{C}{K_m K_f} = \frac{1}{5 \times 10^5} = 2\,\mu\text{F}$$

Verify: $\omega_0' = 1/\sqrt{L'C'} = 1/\sqrt{(5 \times 10^{-3})(2 \times 10^{-6})} = 10^4$. ✓And $Q' = \omega_0' L'/R' = (10^4)(0.005)/50 = 1 = Q$, unchanged as expected.

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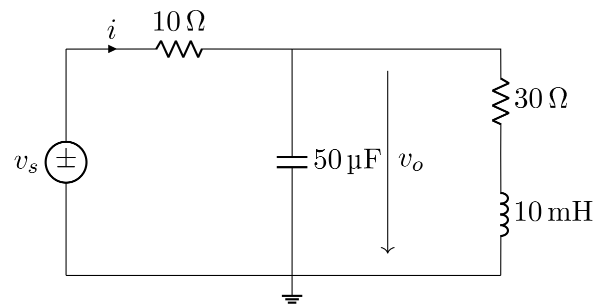
8 Practice Exam

Eight problems, all schematic-based, weighted roughly the way the chapters are. Give yourself 75 minutes, calculator only, no notes beyond §1 or the separate formula sheet. Solutions start on the following page — do not look until you are done.

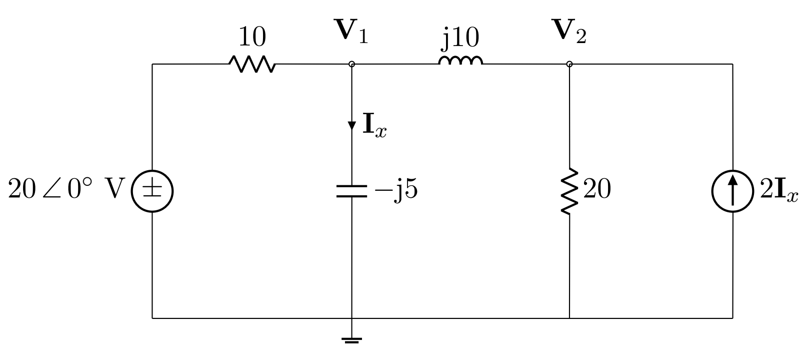
Problem 1 (Ch. 9)

For the circuit below with $v_s(t) = 60\cos(1000t)$ V, find $\mathbf{Z}_{\text{in}}$, $i(t)$, and $v_o(t)$.



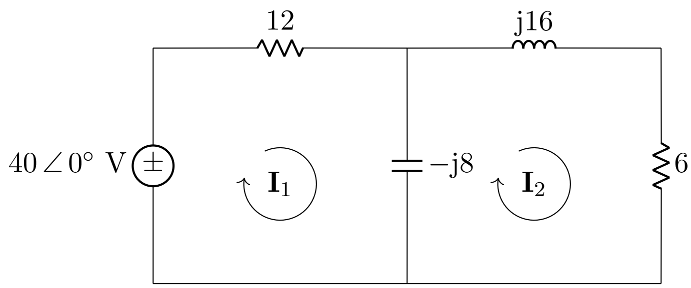
Problem 2 (Ch. 10)

Use nodal analysis to find $\mathbf{V}_1$ and $\mathbf{V}_2$. The dependent source is controlled by $\mathbf{I}_x$, the current flowing *down* through the capacitor.



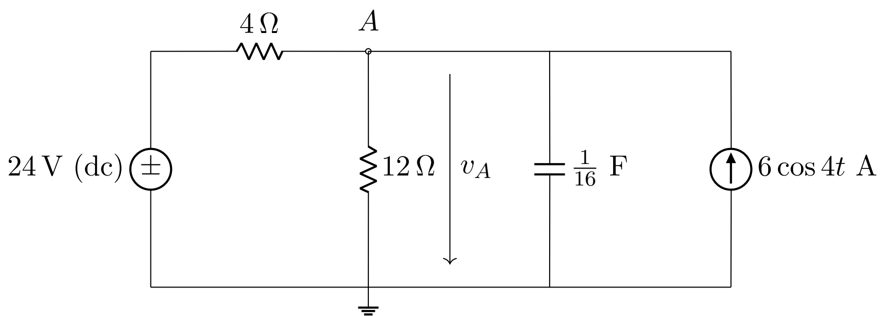
Problem 3 (Ch. 10)

Use mesh analysis to find $\mathbf{I}_1$, $\mathbf{I}_2$, and the current through the capacitor.



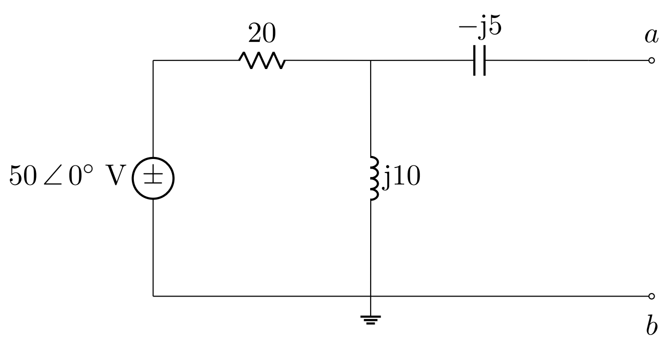
Problem 4 (Ch. 10)

Find $v_A(t)$ using superposition.



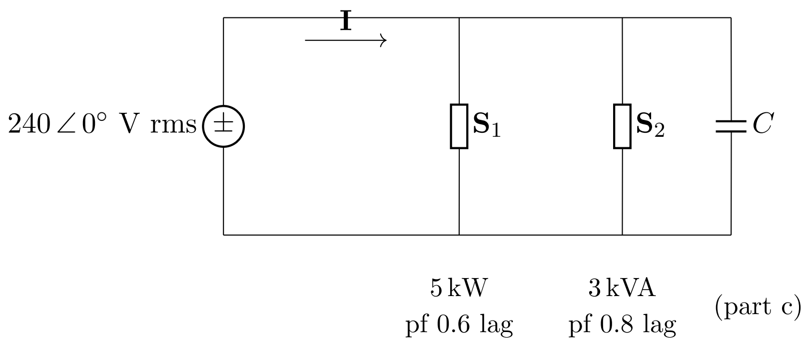
Problem 5 (Ch. 10 and 11)

(a) Find the Thévenin equivalent at a–b. (b) Find the load $\mathbf{Z}_L$ that draws maximum average power and compute that power.



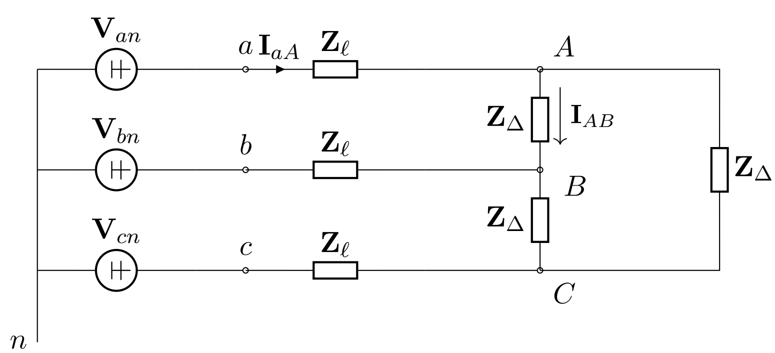
Problem 6 (Ch. 11)

A 240 V rms, 60 Hz line feeds two parallel loads: load 1 absorbs 5 kW at pf 0.6 lagging, load 2 absorbs 3 kVA at pf 0.8 lagging. Find (a) the total complex power and overall pf, (b) the line current, and (c) the parallel capacitance needed to bring the pf to 0.95 lagging, and the new line current.



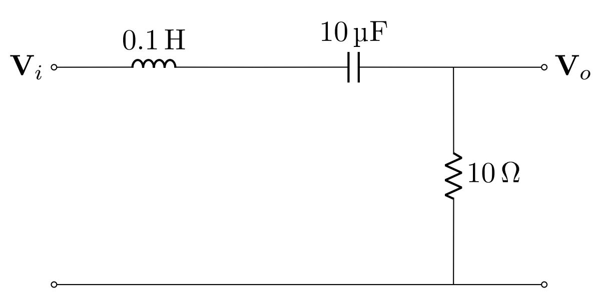
Problem 7 (Ch. 12)

A balanced *abc* Y-connected source with $\mathbf{V}_{an} = 120\angle 0°$ V rms feeds a balanced Δ load of $\mathbf{Z}_\Delta = 21 + \text{j}27\ \Omega$ per phase through lines of impedance $\mathbf{Z}_\ell = 2 + \text{j}3\ \Omega$. Find the line current $\mathbf{I}_{aA}$, the load phase current $\mathbf{I}_{AB}$, the power delivered to the load, and the power lost in the lines.



Problem 8 (Ch. 14)

For the series *RLC* circuit below with the output taken across the resistor: (a) derive $\mathbf{H}(\omega) = \mathbf{V}_o/\mathbf{V}_i$ and identify the filter type; (b) find ω_0, f_0, Q, B, and the half-power frequencies; (c) state $|\mathbf{H}|$ at ω_0 and at ω_1; (d) if the circuit is frequency-scaled by $K_f = 100$ and magnitude-scaled by $K_m = 5$, give the new element values and the new ω_0 and Q.



9 Practice Exam — Answer Key

Problem 1

$\omega = 1000$: capacitor $\to -\mathrm{j}20\ \Omega$, inductor $\to \mathrm{j}10\ \Omega$.

$$\mathbf{Z}_p = \frac{(-\mathrm{j}20)(30+\mathrm{j}10)}{30+\mathrm{j}10-\mathrm{j}20} = \frac{200-\mathrm{j}600}{30-\mathrm{j}10} = 12-\mathrm{j}16 = 20\angle -53.13^\circ\ \Omega$$

$$\mathbf{Z}_{\text{in}} = 10 + (12-\mathrm{j}16) = 22-\mathrm{j}16\ \Omega = 27.20\angle -36.03^\circ\ \Omega$$

$$\mathbf{I} = \frac{60\angle 0^\circ}{27.20\angle -36.03^\circ} = 2.206\angle 36.03^\circ\ \text{A} \qquad \mathbf{V}_o = \mathbf{I}\mathbf{Z}_p = 44.11\angle -17.10^\circ\ \text{V}$$

$$i(t) = 2.206\cos(1000t + 36.03^\circ)\ \text{A} \qquad v_o(t) = 44.11\cos(1000t - 17.10^\circ)\ \text{V}$$

Problem 2

Node 1: $\dfrac{20-\mathbf{V}_1}{10} = \dfrac{\mathbf{V}_1}{-\mathrm{j}5} + \dfrac{\mathbf{V}_1-\mathbf{V}_2}{\mathrm{j}10}$

Node 2: $\dfrac{\mathbf{V}_1-\mathbf{V}_2}{\mathrm{j}10} + 2\mathbf{I}_x = \dfrac{\mathbf{V}_2}{20}$, with $\mathbf{I}_x = \dfrac{\mathbf{V}_1}{-\mathrm{j}5}$

In matrix form:

$$\begin{bmatrix} 0.1+\mathrm{j}0.1 & \mathrm{j}0.1 \\ \mathrm{j}0.5 & -0.05+\mathrm{j}0.1 \end{bmatrix} \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$\boxed{\mathbf{V}_1 = 14.14\angle 98.13^\circ\ \text{V}, \qquad \mathbf{V}_2 = 37.95\angle -108.44^\circ\ \text{V}}$$

and $\mathbf{I}_x = \mathbf{V}_1/(-\mathrm{j}5) = 2.828\angle -171.87^\circ$ A.

Problem 3

$$\begin{bmatrix} 12-\mathrm{j}8 & \mathrm{j}8 \\ \mathrm{j}8 & 6+\mathrm{j}8 \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} 40 \\ 0 \end{bmatrix}$$

(The off-diagonal terms are $+\mathrm{j}8$ because the shared impedance is $-\mathrm{j}8$ and it enters with a minus sign.)

$$\boxed{\mathbf{I}_1 = 1.945\angle 39.63^\circ\ \text{A}, \qquad \mathbf{I}_2 = 1.556\angle -103.50^\circ\ \text{A}}$$

$$\mathbf{I}_C = \mathbf{I}_1 - \mathbf{I}_2 = 3.323\angle 55.95^\circ\ \text{A}$$

Problem 4

DC alone (current source open, capacitor open): $v_{A1} = 24 \cdot \frac{12}{4+12} = 18\,\text{V}$.

AC alone ($\omega = 4$, voltage source shorted): $1/\mathrm{j}\omega C = 1/(\mathrm{j}\cdot 4 \cdot \frac{1}{16}) = -\mathrm{j}4\ \Omega$. The $4\,\Omega$, $12\,\Omega$, and $-\mathrm{j}4$ are all in parallel across the current source:

$$\mathbf{Y} = 0.25 + 0.0833 + \mathrm{j}0.25 = 0.4167\angle 36.87^\circ\ \text{S} \qquad \mathbf{V}_{A2} = \frac{6}{\mathbf{Y}} = 14.4\angle -36.87^\circ\ \text{V}$$

$$\boxed{v_A(t) = 18 + 14.4\cos(4t - 36.87^\circ)\ \text{V}}$$

Problem 5

(a) With a–b open, no current in the −j5 capacitor, so $\mathbf{V}_{\text{Th}}$ is the inductor voltage:

$$\mathbf{V}_{\text{Th}} = 50\angle 0^\circ \cdot \frac{\text{j}10}{20+\text{j}10} = 22.36\angle 63.43^\circ \text{ V}$$

Shorting the source, $20 \parallel \text{j}10 = 4+\text{j}8$, then add the series −j5:

$$\mathbf{Z}_{\text{Th}} = -\text{j}5 + (4+\text{j}8) = 4+\text{j}3\ \Omega = 5\angle 36.87^\circ\ \Omega$$

(b)

$$\boxed{\mathbf{Z}_L = \mathbf{Z}_{\text{Th}}^* = 4-\text{j}3\ \Omega} \qquad P_{\max} = \frac{|\mathbf{V}_{\text{Th}}|^2}{8R_{\text{Th}}} = \frac{500}{32} = \boxed{15.625\,\text{W}}$$

Problem 6

(a) $\theta_1 = 53.13^\circ$, $S_1 = 5/0.6 = 8.333\,\text{kVA}$: $\mathbf{S}_1 = 5+\text{j}6.667$ kVA.
$\theta_2 = 36.87^\circ$, $S_2 = 3\,\text{kVA}$: $\mathbf{S}_2 = 2.4+\text{j}1.8$ kVA.

$$\mathbf{S}_T = 7.4+\text{j}8.467 \text{ kVA} = 11.24\angle 48.85^\circ \text{ kVA}, \qquad \text{pf} = \cos 48.85^\circ = 0.658 \text{ lagging}$$

(b) $I = S_T/V = 11{,}245/240 = 46.85\,\text{A rms}$.

(c) $Q_2 = P\tan(18.19^\circ) = 7400(0.3287) = 2432\,\text{VAR}$, so $Q_C = 8467 - 2432 = 6034\,\text{VAR}$.

$$C = \frac{6034}{2\pi(60)(240)^2} = \boxed{278\,\mu\text{F}} \qquad I_{\text{new}} = \frac{7400/0.95}{240} = 32.5\,\text{A}$$

A 31 % current reduction with P unchanged.

Problem 7

Convert the Δ load to Y: $\mathbf{Z}_Y = (21+\text{j}27)/3 = 7+\text{j}9\ \Omega$. Per phase,

$$\mathbf{Z}_{\text{total}} = (2+\text{j}3) + (7+\text{j}9) = 9+\text{j}12 = 15\angle 53.13^\circ\ \Omega$$

$$\mathbf{I}_{aA} = \frac{120\angle 0^\circ}{15\angle 53.13^\circ} = \boxed{8\angle -53.13^\circ \text{ A}}$$

Back out the Δ phase current (divide by $\sqrt{3}$, advance 30°):

$$\mathbf{I}_{AB} = \frac{8}{\sqrt{3}}\angle(-53.13^\circ + 30^\circ) = \boxed{4.619\angle -23.13^\circ \text{ A}}$$

$$P_{\text{load}} = 3|\mathbf{I}_{AB}|^2(21) = 3(21.33)(21) = 1344\,\text{W}$$
$$P_{\text{line}} = 3|\mathbf{I}_{aA}|^2(2) = 3(64)(2) = 384\,\text{W}$$

Check: total supplied $= \sqrt{3}V_L I_L\cos\theta = \sqrt{3}(207.8)(8)\cos 53.13^\circ = 1728\,\text{W} = 1344 + 384$. ✓

Problem 8

(a) Voltage divider with the output across R:

$$\mathbf{H}(\omega) = \frac{R}{R + \mathrm{j}\omega L + \frac{1}{\mathrm{j}\omega C}} = \frac{\mathrm{j}\omega RC}{1 - \omega^2 LC + \mathrm{j}\omega RC}$$

As $\omega \to 0$, $\mathbf{H} \to 0$; as $\omega \to \infty$, $\mathbf{H} \to 0$; and at ω_0 the reactances cancel so $\mathbf{H} = 1$. This is a **band-pass** filter.

(b)

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(0.1)(10^{-5})}} = 1000\,\mathrm{rad\,s^{-1}}, \qquad f_0 = \frac{1000}{2\pi} = 159.2\,\mathrm{Hz}$$

$$Q = \frac{\omega_0 L}{R} = \frac{(1000)(0.1)}{10} = 10, \qquad B = \frac{R}{L} = 100\,\mathrm{rad\,s^{-1}}$$

$Q = 10$ is right at the high-Q threshold, so $\omega_1 \approx 950$, $\omega_2 \approx 1050\,\mathrm{rad\,s^{-1}}$.

(c) $|\mathbf{H}(\omega_0)| = 1$ (0 dB); $|\mathbf{H}(\omega_1)| = 1/\sqrt{2} = 0.707$ (−3 dB).

(d) With $K_m = 5$ and $K_f = 100$:

$$R' = 5(10) = 50\,\Omega, \qquad L' = \frac{5}{100}(0.1) = 5\,\mathrm{mH}, \qquad C' = \frac{10^{-5}}{(5)(100)} = 0.02\,\mu\mathrm{F}$$

$$\omega_0' = K_f\omega_0 = 100\,\mathrm{krad\,s^{-1}}, \qquad Q' = Q = 10 \text{ (unchanged)}, \qquad B' = 10\,\mathrm{krad\,s^{-1}}$$

10 Last-Night Checklist

Things that cost points every time:

1. **Not converting to cosine first.** $\sin(\omega t) = \cos(\omega t - 90°)$. Every sine in the problem statement is a trap.
2. **Forgetting the $\frac{1}{2}$** in $P = \frac{1}{2} V_m I_m \cos\theta$ when the phasors are amplitudes, or including it when they are already RMS.
3. **Using the angle of $\mathbf{I}_L$ instead of the angle of $\mathbf{Z}$** for θ in three-phase power. θ is always the load impedance angle.
4. **Adding apparent powers.** Add $P + \mathrm{j}Q$, then take the magnitude.
5. **Using $1/\sqrt{LC}$ for a practical parallel resonant circuit** that has resistance in series with the inductor. Use $\omega_0 = \sqrt{1/LC - R^2/L^2}$.
6. **Superposing phasors from different frequencies.** Add the time functions, not the phasors.
7. **Sign of the off-diagonal terms** in the mesh matrix: the shared impedance always enters with a minus sign, which flips the visible sign when the impedance itself is negative imaginary.
8. **Leaving ω out of the final time-domain answer.** Write $\cos(\omega t + \phi)$ with the actual numeric ω, not just the phasor.
9. **$\sqrt{3}$ in the wrong place.** Y: currents match, voltages scale. Δ: voltages match, currents scale.
10. **Bode plots not in standard form.** $(2 + \mathrm{j}\omega)$ is not a corner factor; $2(1 + \mathrm{j}\omega/2)$ is.

Five-minute self-test. If you can answer these without notes you are in good shape:

1. What is the impedance of a 2 µF capacitor at 5 kHz?
2. A load draws 10 kW at pf 0.5 lagging. What is Q?
3. In a balanced Δ load with $I_p = 10\,\mathrm{A}$, what is I_L and what is the phase relationship?
4. A series RLC has $\omega_0 = 10\,\mathrm{krad\,s^{-1}}$ and $B = 500\,\mathrm{rad\,s^{-1}}$. What is Q? Is the symmetric approximation for ω_1, ω_2 valid?
5. $\mathbf{H}(s) = \dfrac{50}{s(s+25)}$. What is the low-frequency slope and where is the corner?

Answers: (1) $-\mathrm{j}15.92\ \Omega$. (2) $Q = 10\tan(60°) = 17.32\,\mathrm{kVAR}$. (3) $I_L = \sqrt{3}(10) = 17.32\,\mathrm{A}$, lagging $\mathbf{I}_p$ by 30°. (4) $Q = 20$, yes, high-Q. (5) Standard form is $\dfrac{2}{\mathrm{j}\omega(1 + \mathrm{j}\omega/25)}$: slope -20 dB/dec at low ω, corner at $\omega = 25$, then -40 dB/dec.